



NORTH SYDNEY BOYS' HIGH SCHOOL
2008 Preliminary Course Assessment Task 2

MATHEMATICS

General instructions

- Working time – 60 minutes.
- Write in the booklet provided.
- Each new question is to be started on a new page.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- Question are **not** of equal value.
- At the conclusion of the examination, bundle the booklet within this paper and hand to examination supervisors.

Total Marks: 64

NAME:

Marker's use only.

QUESTION	MARKS
1	/14
2	/28
3	/11
4	/11
Total	/64
Total (%)	/100

Question 1 (14 Marks)		Commence a new page.	Marks
(a)	Express $0.\dot{6}\dot{1}$ as a fraction in simplest form.		2
(b)	A metal cube has volume of 157.464 m^3 . Find its surface area.		2
(c)	Factorise $125a^3 - 8b^3$.		2
(d)	Solve:		
	i. $\frac{x}{2} + 7 = 9$.		2
	ii. $y(2y + 5) = 12$.		3
(e)	Sketch the curve $y = (x + 5)(4x - 3)$. Hence or otherwise, solve		3
	$(x + 5)(4x - 3) \geq 0$		

Question 2 (28 Marks)		Commence a new page.	
- Sketch the following on separate number planes, showing any intercepts with axes or asymptotes. - State the domain & range for each function.			
(a)	$y = -2x + 5$.		4
(b)	$g(x) = 1 - x $.		4
(c)	$f(x) = 16 - x^2$.		4
(d)	$f(x) = \sqrt{16 - x^2}$.		4
(e)	$y = 4^{-x}$.		4
(f)	$y = \sqrt{x - 1}$.		4
(g)	$y = -\frac{1}{x - 2}$.		4

Question 3 (11 Marks)		Commence a new page.	Marks
(a)	Show whether the following functions are odd, even or neither:		
i.	$y = x^2 - 1$.		2
ii.	$y = x^5 - 2x^3$.		2
iii.	$y = 2x + 1$.		2
iv.	$y = 2x + 1$.		2
(b)	Solve the inequality $ 2x - 1 > 6$ and graph the solution on the number line.		3

Question 4 (11 Marks)		Commence a new page.	Marks
(a)	Given $f(x) = x^2 - 3$, evaluate:		
i.	$f(2)$.		1
ii.	$f(3) - f(0)$.		1
iii.	$f(a + 1)$		2
(b)	Sketch the following function:		3

$$f(x) = \begin{cases} -x & x \geq 2 \\ -1 & x < 2 \end{cases}$$

(c)	For the function $y = \frac{x^2 - 4}{x + 2}$		
i.	Find its domain.		1
ii.	By simplifying the function, find its range.		1
iii.	Sketch the function.		2

End of paper.

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Solutions

Question 1

(a) (2 marks)

$$\begin{array}{r} x = 0.6\dot{1} \\ 10x = 6.1\dot{1} \\ \hline \therefore 9x = 5.5 \quad [1] \\ x = \frac{55}{90} = \frac{11}{18} \quad [1] \end{array}$$

(b) (2 marks)

$$V = 157.464 \text{ m}^3$$

Letting x be the side length of the square,

$$\begin{aligned} x^3 &= 157.464 \\ x &= \frac{27}{5} \quad [1] \\ SA &= 6x^2 \\ &= 6 \times \left(\frac{27}{5}\right)^2 \\ &= 174.96 \text{ m}^2 \quad [1] \end{aligned}$$

(c) (2 marks)

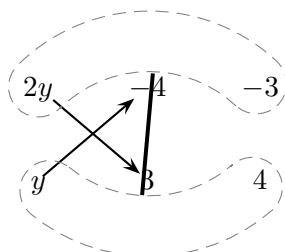
$$125a^3 - 8b^3 = \overbrace{(5a - 2b)}^{[1]} \overbrace{(25a^2 + 10ab + 4b^2)}^{[1]}$$

(d) i. (2 marks)

$$\begin{aligned} \frac{x}{2} + \frac{7}{-7} &= \frac{9}{-7} \\ \frac{x}{2} &= 2 \quad [1] \\ x &= 4 \quad [1] \end{aligned}$$

ii. (3 marks)

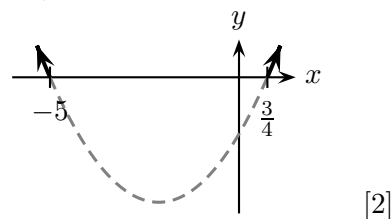
$$\begin{aligned} y(2y + 5)_{-12} &= \frac{12}{-12} \\ 2y^2 + 5y - 12 &= 0 \quad [1] \end{aligned}$$



$$(2y - 3)(y + 4) = 0 \quad [1]$$

$$\therefore y = \frac{3}{2}, -4 \quad [1]$$

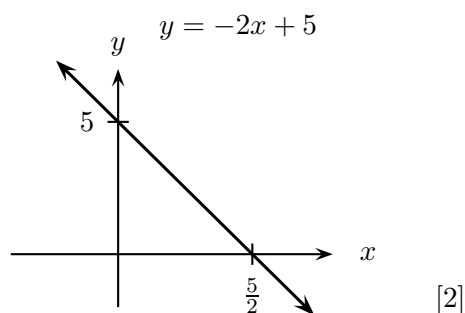
(e) (3 marks)



$$x < -5 \quad \text{or} \quad x > \frac{3}{4} \quad [1]$$

Question 2

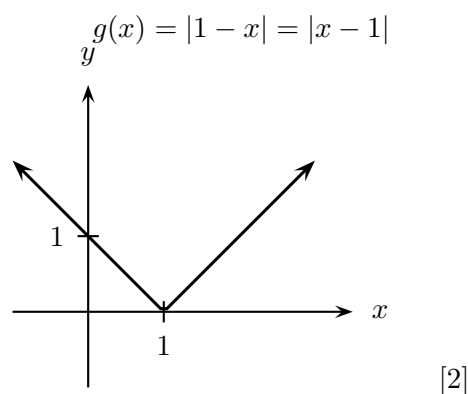
(a) (4 marks)



$$D = \{x : x \in \mathbb{R}\} \quad [1]$$

$$R = \{y : y \in \mathbb{R}\} \quad [1]$$

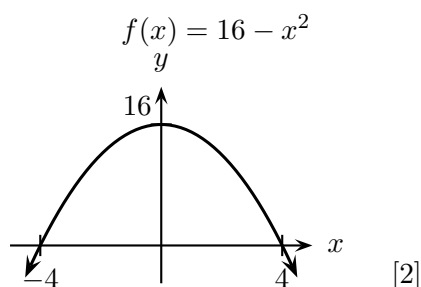
(b) (4 marks)



$$D = \{x : x \in \mathbb{R}\} \quad [1]$$

$$R = \{y : y \geq 0\} \quad [1]$$

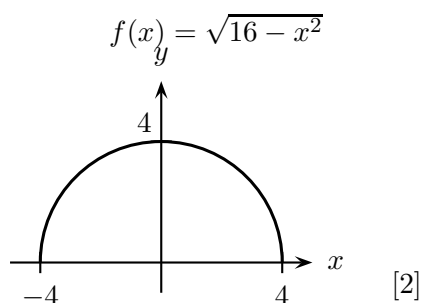
(c) (4 marks)



$$D = \{x : x \in \mathbb{R}\} \quad [1]$$

$$R = \{y : y \leq 16\} \quad [1]$$

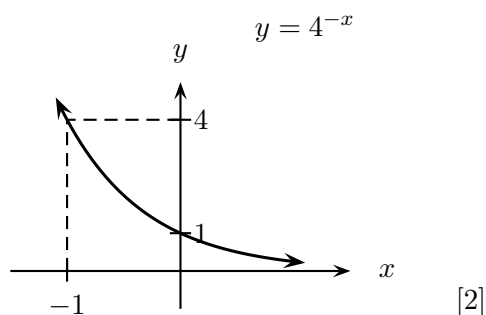
(d) (4 marks)



$$D = \{x : -4 \leq x \leq 4\} \quad [1]$$

$$R = \{y : 0 \leq y \leq 4\} \quad [1]$$

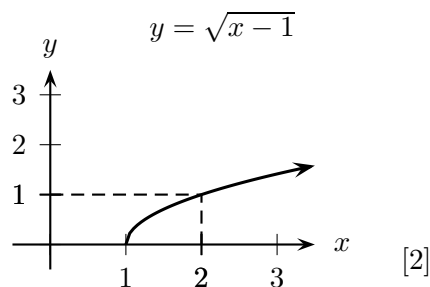
(e) (4 marks)



$$D = \{x : x \in \mathbb{R}\} \quad [1]$$

$$R = \{y : y > 0\} \quad [1]$$

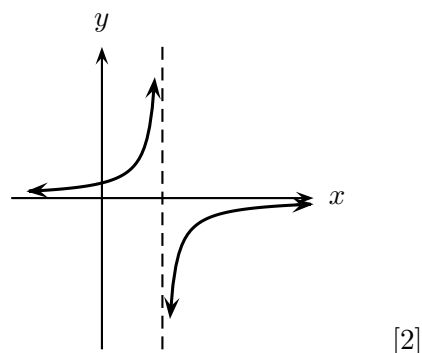
(f) (4 marks)



$$D = \{x : x \geq 1\} \quad [1]$$

$$R = \{y : y \geq 0\} \quad [1]$$

(g) (4 marks)



$$D = \{x : x \neq 2\} \quad [1]$$

$$R = \{y : y \neq 0\} \quad [1]$$

Question 3

(a) i. (2 marks)

$$f(x) = x^2 - 1$$

$$f(a) = a^2 - 1$$

$$f(-a) = (-a)^2 - 1 = f(a)$$

$$= a^2 - 1$$

$$\therefore f(x) = x^2 - 1 \text{ is even.}$$

Alternatively, a sketch will suffice.

ii. (2 marks)

$$f(x) = x^5 - 2x^3$$

$$f(-a) = (-a)^5 - 2(-a)^3$$

$$= -a^5 + 2a^3$$

$$= -(a^5 - 2a^3) = -f(a)$$

$$f(-a) = -f(a)$$

$$\therefore f(x) = x^5 - 2x^3 \text{ is odd.}$$

iii. (2 marks)

$$\begin{aligned}
 f(x) &= 2x + 1 \\
 f(-a) &= 2(-a) + 1 \\
 &= -2a + 1 \\
 -f(a) &= -2a - 1 \\
 f(a) &= 2a + 1 \\
 f(a) &\neq f(-a) \neq -f(a) \\
 \therefore f(x) &\text{ is neither odd or even}
 \end{aligned}$$

iv. (2 marks)

$$\begin{aligned}
 f(x) &= |2x| + 1 \\
 f(a) &= |2a| + 1 \\
 &= 2|a| + 1 \\
 f(-a) &= |2(-a)| + 1 \\
 &= |-2a| + 1 \\
 &= 2|a| + 1 \\
 f(a) &= f(-a) \\
 \therefore f(x) &\text{ is even.}
 \end{aligned}$$

Again a graph would suffice. It is the absolute value graph $f(x) = |2x|$ shifted up by 1 unit.

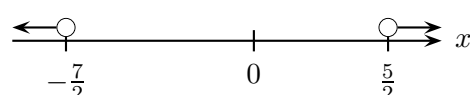
(b) (3 marks)

$$\begin{aligned}
 |2x - 1| &> 6 \\
 |2x - 1| &= \begin{cases} 2x - 1 & \text{if } 2x - 1 > 0 \\ -(2x - 1) & \text{if } 2x - 1 < 0 \end{cases} \quad [1]
 \end{aligned}$$

Case 1: $2x - 1 > 0$

$$\begin{aligned}
 2x - 1 &> 6 \\
 +1 &+1 \\
 2x &> 7 \\
 \div 2 &\div 2 \\
 x &> \frac{7}{2} \quad [1]
 \end{aligned}$$

Case 2: $2x - 1 < 0$

$$\begin{aligned}
 -(2x - 1) &> 6 \\
 -2x + 1 &> 6 \\
 -1 &-1 \\
 -2x &> 5 \\
 \div -2 &\div -2 \\
 x &< -\frac{5}{2} \\
 \therefore x &> \frac{7}{2} \quad x < -\frac{5}{2} \quad [1]
 \end{aligned}$$


Question 4

(a) i. (1 mark)

$$\begin{aligned}
 f(2) &= 2^2 - 3 \\
 &= 1
 \end{aligned}$$

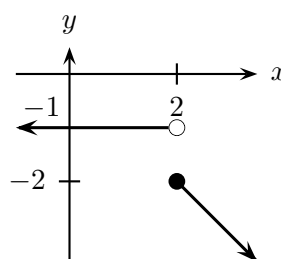
ii. (1 mark)

$$\begin{aligned}
 f(3) - f(0) &= (3^2 - 3) - (0 - 3) \\
 &= 9
 \end{aligned}$$

iii. (2 marks)

$$\begin{aligned}
 f(a + 1) &= (a + 1)^2 - 3 \quad [1] \\
 &= a^2 + 2a + 1 - 3 \\
 &= a^2 + 2a - 2 \quad [1]
 \end{aligned}$$

(b) (3 marks)



- [1] for sketching $f(x) = -x$ for $x \geq 2$.
- [1] for sketching $f(x) = -1$ for $x < 2$.
- [1] for correct open circle & closed circles at $x = 2$.

(c) i. (1 mark)

$$D = \{x : x \neq -2\}$$

ii. (1 mark)

$$\begin{aligned}
 y &= \frac{(x - 2)(\cancel{x + 2})}{\cancel{(x + 2)}} \\
 &= x - 2 \quad x \neq -2 \\
 R &= \{y : y \neq -4\}
 \end{aligned}$$

iii. (2 marks)

